Table 4 Overview of economic studies reporting health state utility values

Author, year	Region	Study type	Patient population	Treatment	Utility values on-treatment	Utility values in remission	Utility values for disease progression
Li 2022 [54]	China	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Axi-cel Salvage chemotherapy	Disutility of chemotherapy: − 0.42[a]	0.83	0.39
Wakase 2021 [55]	Japan	CUA	Adult patients with CD19-positive R/R DLBCL who are ineligible for, or relapsed after, autologous HSCT	Tisagenlecleucel Salvage chemotherapy	− 0.15; 28 days[b] − 0.15; 62 days[b]	0.83	0.39
Cher 2020 [53]	Singapore	CUA	Adult patients with R/R DLBCL from JULIET trial and CORAL extension study	Tisagenlecleucel Salvage chemotherapy	Disutility: − 0.15[c] Disutility: − 0.15[c]	0.7	0.59
Wang 2021 [58]	Singapore	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Tisagenlecleucel Salvage chemotherapy	Disutility: − 0.15[d] Disutility: − 0.15[d]	0.90 (SE, 0.01) 0.90 (SE, 0.01)	0.82 (SE, 0.02) 0.82 (SE, 0.02)
Bastos-Oreiro 2022 [71]	Spain	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Axi-cel Tisagenlecleucel	Month 1: 0.740	≤ 12 months: 0.782 > 12 months: 0.820	0.39
Muszbek 2016 [62]	UK	CUA	Patients with R/R aggressive NHL receiving third- or fourth-line treatment	Pixantrone	Grade 2 AE disutility: 0.0075 Grade 3/4 AE disutility: 0.0078	Preprogression: 0.76 (SE, 0.03)	0.68 (SE, 0.04)
				Current clinical practice	Grade 2 AE disutility: 0.0066 Grade 3/4 AE disutility: 0.0073		
Betts 2020 [61] Betts 2020 [72]	US	CEA	Adult patients with R/R DLBCL based on GO29365 trial	Pola-BR BR	NR	0.83	0.71
Liu 2021 [49]	US	CUA	Adult patients with R/R DLBCL with ≥ 2 prior lines of systemic therapies	Axi-cel Tisagenlecleucel	Month 1: 0.740	First 24 months: 0.782 > 24 months since: 0.820	0.39
Lin 2019 [50]	US	CUA	US adults with R/R DLBCL after ≥ 2 lines of therapy or relapsed ≤ 12 months after HSCT	Axi-cel	Months 1–2: 0.50	0.70	0.45
				Tisagenlecleucel	Months 1–2: 0.58	0.70	
				Chemoimmunotherapy	During treatment: 0.63	0.71	
				Autologous HSCT	Months 1–2: 0.43 Month 3 (if in remission): 0.70	0.70	

Table 4 (continued)

Author, year	Region	Study type	Patient population	Treatment	Utility values on-treatment	Utility values in remission	Utility values for disease progression
Oluwole 2022 [48]	US	CUA	US adults with R/R LBCL after ≥ 2 lines of therapy	Axi-cel Liso-cel	First model cycle: 0.740	≤ 24 months since model entry: 0.782 > 24 months since model entry: 0.820	0.390
Roth 2018 [51]	US	CUA	Adult patients with R/R DLBCL based on ZUMA-1 trial	Axi-cel Salvage chemotherapy	0.740 0.673	< 6-month follow-up: 0.782 ≥ 6-month follow-up: 0.823	0.390
Whittington 2019 [56]	US	CUA	Adult patients with R/R B cell lymphoma from ZUMA-1 trial	Axi-cel Chemotherapy HSCT	NR Disutility: − 0.42 Disutility: − 0.57	0.83	0.39
Calamia 2021 [59]	US	CUA	Adult patients with R/R DLBCL who are transplant ineligible based on L-MIND and G029365 trials	Pola-BR Tafasitamab + lenalidomide	AE disutility: − 0.02 AE disutility: − 0.01	PFS: 0.83 PFS adjusted for disutility: 0.82	0.39
Patel 2020 [60]	US	CUA	Adult patients with R/R follicular or DLBCL (mirrored phase 2 cohort from NCT02257567)	Pola-BR BR	Disutility from chemotherapy: − 0.42	0.83	0.39
Qi 2021 [57]	US	CUA	Adults with R/R DLBCL after ≥ 2 lines of therapy	Tisagenlecleucel Salvage chemotherapy	Disutility: − 0.15 (duration: 27.9 days)[e] Disutility: − 0.15 (duration: 72.2 days)[e]	0.83	0.39

AE adverse event; *axi-cel* axicabtagene ciloleucel; *BR* bendamustine and rituximab; *CRS* cytokine release syndrome; *CEA* cost-effectiveness analysis; *CUA* cost-utility analysis; *DLBCL* diffuse large B cell lymphoma; *HSCT* hematopoietic stem cell transplantation; *ICU* intensive care unit; *NHL* non-Hodgkin lymphoma; *NR* not reported; *PFS* progression-free survival; *pola-BR* polatuzumab vedotin, bendamustine, and rituximab; *R/R* relapsed or refractory; *SE* standard error; *UK* United Kingdom; *US* United States

[a]Reports additional disutility value for HSCT (− 0.57)

[b]Reports additional disutility value for subsequent HSCT (− 0.30/year)

[c]Disutilities were also provided for ICU stay for CRS (− 0.7), AEs (− 0.15), and HSCT (− 0.15)

[d]Disutilities were also provided for ICU stay for CRS (− 0.9), ICU stay for non-CRS AEs (− 0.9), and subsequent HSCT (− 0.30/year)

[e]Disutilities were also provided for ICU stay for CRS (− 0.83; duration 8.5 days) and ICU stay for non-CRS AEs (− 0.83; duration 0.9 days) for tisagenlecleucel-infused patient, and subsequent HSCT (− 0.30/year) for both treatments